

A Low-Complexity Codebook Design for Large-Scale SCMA

Chao Yang^{1,2,3,4}, Shuseng Jing^{1,2,3,4}, Xiao Liang^{1,2,3,4}, Zaichen Zhang^{2,3,4}, Xiaohu You^{2,3}, and Chuan Zhang^{1,2,3,4,*}

¹Lab of Efficient Architectures for Digital-communication and Signal-processing (LEADS)

²National Mobile Communications Research Laboratory, Southeast University

³School of Information Science and Engineering, Southeast University

⁴Quantum Information Center of Southeast University, Nanjing, China

Email: {chaoyang, chzhang}@seu.edu.cn

Abstract—Sparse-code multiple-access (SCMA) is an effective non-orthogonal multiple-access (NOMA) technique, which ranks one of the most promising candidates for future wireless. However, most of the existing work prefer a low-connectivity SCMA system because of the difficulties in designing proper codebook, which limits its priorities for massive connection. In this paper, we propose a systematic, low-complexity codebook design for large-scale SCMA. Stage-based generation with quadrature amplitude modulation (QAM) is adopted in the codebook design for SCMA system with arbitrary parameters, where the layer operator of each user is fixed as constellation rotation for simplification. Extension from known codebook is another proposed design manner, which allocates the known codeword repetitively based on specific allocation order. With a effectiveness metric, numerical results reveal advantages of our work. Hardware architecture of the detector is also proposed in this paper.

Index Terms—Sparse-code multiple-access (SCMA), massive connection, constellation rotation, extension, hardware.

I. INTRODUCTION

Sparse-code multiple-access (SCMA) is regarded as one promising multiple-access (MA) technique, which interests applications like 5G because of its good overloading tolerance [1]. However, existing SCMA systems prefer a low-connectivity factor graph, which limits its applications. The main reason is the difficulty in designing effective codebook with low complexity for large-scale SCMA systems [2].

For codebook design, related approaches have been proposed, which involve constellation transformation of arbitrary inner connection [3], still lack operability or bring much complexity. A stage-based codebook design is proposed in [4], but the choice of mother constellation and layer operators requires precise calculation and rich experiences, which makes it only applicable for small-size SCMA. Therefore, a simplified systematic codebook design manner and implementation facing large-scale customers is the must for SCMA application.

In this paper, more practical codebook designs for massive-connection SCMA are proposed. Rotation-only layer operations based on QAM is adopted in codebook design [5], which is evolved from the stage-based manner in [4], providing a simpler manner for constellation construction. In this manner, rotation operations of equal angle (can be unequal under certain conditions) are applied in separated constellation

“cycles” (dimension) with different radius lengths, and gray rules are introduced in the bits allocation of each independent layer. Extension from known codebook is another method we proposed, and its design complexity can be much lower than that of rotation construction method. To demonstrate the effectiveness of our manners, a defined performance function is applied for analysis. A general hardware architecture of the detector for large-scale SCMA is proposed as well.

The remainder of this paper is organized as follows. Section II reviews the preliminaries. Simplified codebook design with rotation-only operations and extension is described in Section III. Section IV gives numerical results. Section V presents the hardware architecture. Section VI concludes the entire paper.

II. PRELIMINARIES

Suppose codeword set, constellation set, and information set are $\mathcal{X}$, $\mathcal{C}$, and $\mathcal{B}$, respectively. Define $\mathbf{x} \in \mathcal{X}$, $\mathbf{c} \in \mathcal{C}$, and $\mathbf{b} \in \mathcal{B}$. $|\mathcal{X}| = K$, $|\mathcal{C}| = N$, and $|\mathcal{B}| = \log_2 M$, where M represents the number of symbols in each codeword set. The first mapping round of SCMA encoding [6, 7] is:

$$g: \mathcal{B} \rightarrow \mathcal{C}, \mathbf{c} = g(\mathbf{b}), \quad (1)$$

where $\mathcal{B} \subset \mathbb{B}^{\log_2 M}$, $\mathcal{C} \subset \mathbb{C}^N$, and g is the constellation mapping function. The second round of mapping is:

$$\mathbf{V}: \mathcal{C} \rightarrow \mathcal{X}, \mathbf{x} = \mathbf{V}\mathbf{c}, \quad (2)$$

where $\mathcal{X} \subset \mathbb{C}^K$, and $\mathbf{V} \in \mathbb{B}^{K \times N}$ is the mapping matrix.

We define $\mathbf{f}_j = \text{diag}(\mathbf{V}_j \mathbf{V}_j^T)$, $j = 1, \dots, J$ and call it the binary indicator vector of layer- j , where J is the total number of layers. Now the factor graph matrix, which decides SCMA system, is defined as $\mathbf{F} = (\mathbf{f}_1, \dots, \mathbf{f}_J)$. Each column and row of $\mathbf{F}$ correspond one resource node and one layer node, respectively. So, layer node j and resource node k are connected if and only if $f_{kj} = 1$. Degree of each resource node is defined as $\mathbf{d}_f = (d_{f1}, \dots, d_{fK})^T = \sum_{j=1}^J \mathbf{f}_j$. For more details the readers can refer to [8].

III. CODEBOOK DESIGN WITH LOW COMPLEXITY

A. Stage-based Design Manner

A suitable codebook for SCMA system has always been challenging. Multi-stage design manner could be a solution. This manner is made up of three main stages [4]:

- 1) Generating mapping matrix of each layer based on the chosen factor graph matrix.
- 2) Selecting a reasonable mother constellation with suitable size and appropriate Euclidean distance.
- 3) Acquiring codebook of each layer based on mapping matrix and specific layer operators.

B. Rotation-only Operator for Simplification

Evolved from the staged-based design manner discussed above, QAM constellation with rotation-only operators is adopted in our design, which massively reduce the complexity. We use a (K, J, N) SCMA system with M symbols to demonstrate the design details. Then, the constellation design of each code-dimension (K dimensions in total) is as follows.

- 1) Determine p constellation points for each connected user to arrange on each “circle”. Then, the number of circles is calculated as $L = M/p$ (L must be an integral), and the number of points to construct QAM on each circle is calculated as $T = M \cdot d_f/L$. T points on the l -th circle with radius r_l ($l = 1, 2, \dots, L$) are uniformly distributed.
- 2) Allocate T points on the first circle with radius of r_1 , then the angle between points is calculates as $\theta = 2\pi/T$. p points of each user also compose a uniformly distributed circle, and the bits of each points are assigned according to the rules of Gray code.
- 3) Allocate T points on rest circles ($2, \dots, L$) according to the first circle, and their radius are enlarged into $3r_1, \dots, (2L - 1)r_1$. Rotate the T -points circles with angle $\theta_2, \dots, \theta_L$ ($0 < \theta_2 < \dots < \theta_L < \pi/2$). New acquired points continue the bits allocation based on circle 1 and still obey the rules of Gray code.

Fig. 1 shows the design procedure on the 1-st dimension of $(4, 6, 2)$ regular SCMA system with 4 symbols. As we can see, there are 2 circles in total, and 6 points belonged to 3 independent users are allocated on each circle. In this figure, parameters are set as $r_1 = 1, r_2 = 3, \theta_2 = \pi/6$. The same processing procedure can be applied on the rest code-dimensions of the system, and the final allocation results are shown in Fig. 2 ($\theta_2 = 0$ is the best in this case).

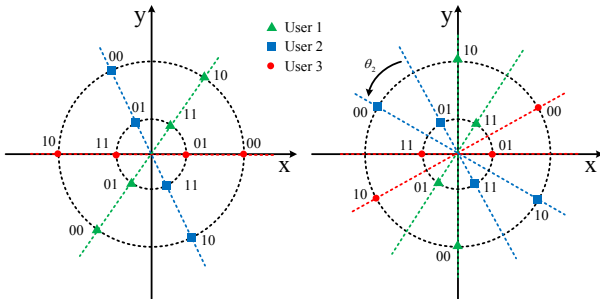


Fig. 1. Constellation rotation procedure on 1-st dimension.

C. Extension from Known Codebook

For further simplification, we can acquire codebook of massive-connection SCMA systems from smaller ones. Suppose the constellation vector of the low-connection SCMA is

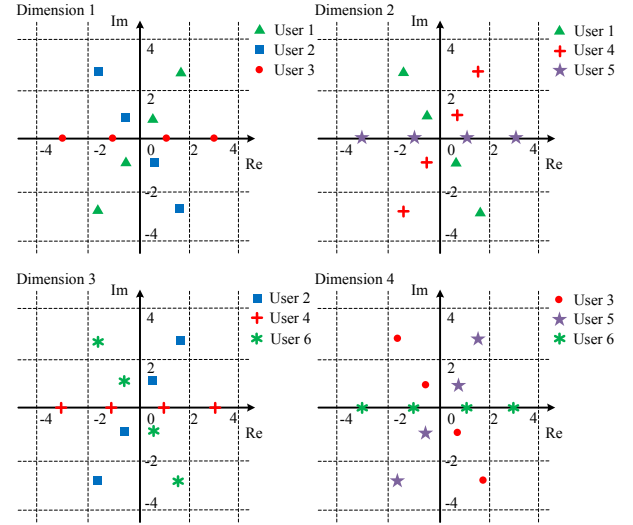


Fig. 2. Constellation allocation of all dimensions.

$\mathbf{c}_l$ ($N_l \times 1$), the acquired constellation vector is $\mathbf{c}_m$ ($N_m \times 1$), and the defined enlarging matrix is $\mathbf{E}_{m,l}$ ($N_m \times N_l$), then the codeword of the new system can be calculated as:

$$\mathbf{x}_m = \mathbf{V}_m \mathbf{c}_m = \mathbf{V}_m \mathbf{E}_{m,l} \mathbf{c}_l \quad (3)$$

where $\mathbf{V}_m$ denotes the mapping matrix of the new system, and $\mathbf{x}_m$ denotes the acquired codeword.

For example, we want to acquired the codebook of $(K = 12, J = 18, N_m = 3)$ SCMA system from the $(K = 4, J = 6, N_l = 2)$ system, then the codeword of user 1 can be transformed as follows.

$$[x_1, x_2, 0, x_4, \mathbf{0}^T]^T = [\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_4][\mathbf{e}'_1, \mathbf{e}'_2, \mathbf{e}'_1]^T [c_1, c_2]^T \quad (4)$$

where $\mathbf{0}$ is the zero vector of (8×1) , $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_4$ are the unit vectors of (12×1) , and $\mathbf{e}'_1, \mathbf{e}'_2$ are the unit vectors of (2×1) . In fact, the enlarging matrix is not necessarily the same for independent layers. The acquired codeword should avoid repetition on the shared resource nodes when applying the enlarging matrix, or it will greatly lower the performance of the decoding. This restriction makes the enlarging matrix of each user normally different for given factor graph matrix, and the allocation of the enlarged codewords (or the determination of enlarging matrixes) should obey the “allocation graph”, which is one possible non-repetition allocation scheme. Fig. 3 shows the allocation graph of $(K = 12, J = 18, N = 3)$ SCMA system as an example. The extension requires 3 copies of the original 6-user codebook to construct the new 18-user one. In Fig. 3, different color represents corresponding layer codebook of the original 6-user system, and numbers “1” and “2” represent the sequence of the non-zero elements in each codeword. As we can see in this figure, all the elements are different in the same row (same number with different colors also means different elements), which ensures the restriction discussed above.

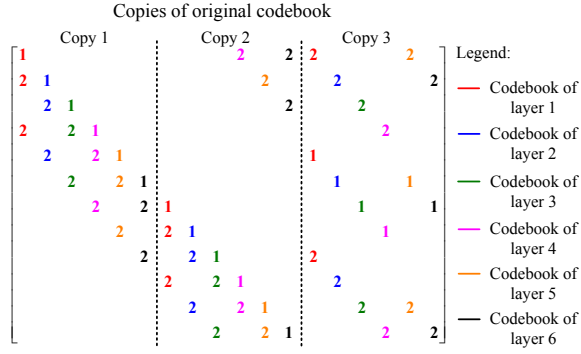


Fig. 3. The allocation graph of “ $K = 12, J = 18, N = 3$ ” SCMA system based on “ $K = 4, J = 6, N = 2$ ” SCMA system.

IV. NUMERICAL RESULTS AND ANALYSIS

A. BER Performance

Fig. 4 shows the BER performance of SCMA systems with different iterations. The method for decoding is deterministic message passing algorithm (DMPA), the codebook size of each user for both systems is 4 (4 symbols to represent “00”, “01”, “10” and “11”), and the channel for simulation is assumed as additional white Gaussian noise (AWGN) channel. We can achieve the following conclusions based on the results.

- 1) Large-scale system with designed manner have performance loss compared to small-scale system with conventional manner, and the smallest SNR gap is about 0.6 dB at $\text{BER} = 10^{-3}$ with constellation rotation design.
- 2) Systems with more decoding iterations enjoy better performance, but the improvement is limited when the iteration number is sufficiently large. Shown by numerical results, 3 iterations is a good choice for both.
- 3) Large-scale system with rotation design outperforms the one with extension, and the SNR gap is about 0.3 dB at $\text{BER} = 10^{-3}$. Extension manner narrows the difference between independent users due to repetitive values in the codebook, but its complexity is lower than rotation manner. Results here are reasonable.

B. Complexity

The design complexity as well as the decoding complexity of the SCMA system are analyzed in this section. For design complexity, we mainly discuss the calculating and transforming complexity of the codebook design in Table I. In this table, color red represents the multiplication-level complexity, color blue represents the addition-level complexity, and color green represents the swap-level complexity. d_f is the degree of resource node, and can be calculated as $d_f = NJ/K$. From the table we can see that the total complexity of our proposed system is much smaller than the conventional system’s.

C. Comprehensive Analysis with Evaluation Function

To analyze the system comprehensively, a evaluation function is defined as follows:

$$\eta = \frac{\text{Throughput}}{\text{SNR}_{1\%} \cdot \text{Complexity}} = f(\lambda, J, M, N, p, \text{SNR}_{1\%}) \quad (5)$$

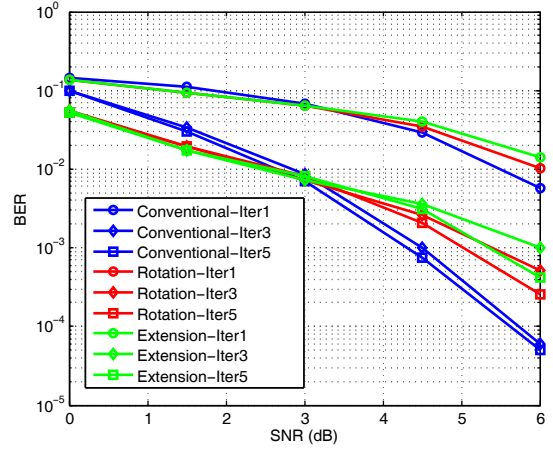


Fig. 4. BER performance of ($K = 4, J = 6, N = 2$) conventional SCMA system and ($K = 12, J = 18, N = 3$) proposed SCMA system with different iterations.

TABLE I
COMPLEXITY ANALYSIS OF SMALL-SCALE SCMA WITH CONVENTIONAL MANNER AND LARGE-SCALE SCMA WITH PROPOSED MANNER.

Size / Manner	Small-scale / Conventional	Large-scale / Proposed
Codebook Complexity	$(M + M^N)N + M^N$	$(M - p)d_f$ or NJ
Decoding Complexity	$Kd_f M^{d_f} + Kd_f [M^{d_f} (d_f + 1) - M + 1] + NMK$	

where $\text{SNR}_{1\%}$ denotes the SNR value to achieve 1% BER performance. The key parameters to influence the comprehensive performance of SCMA system are (K, J, N) or equivalently (λ, J, N), which are actually the parameters to decide the factor graph matrix. To compare the performance in this case, we set ($\lambda = 1.5, M = 4, p = 2$) as constant value for simplification. We will study the performance figure of $N = 2$ and $N = 3$ respectively, as the column weight of the two systems for comparison are “2” and “3” in this paper.

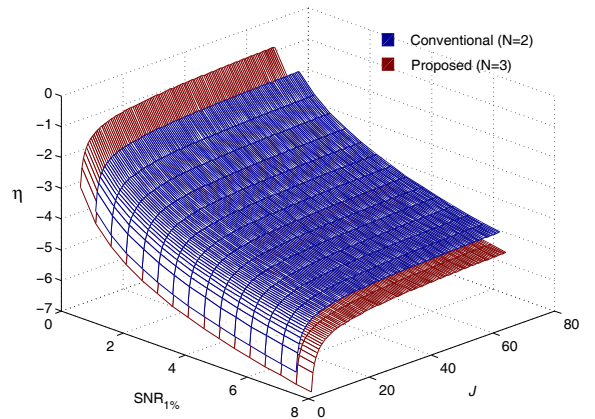


Fig. 5. Evaluation function of conventional and proposed systems.

From Fig. 5, we can see that evaluation function is greatly influenced by parameter J . For a constant $\text{SNR}_{1\%}$, the value

of the function tend to be a constant when J is sufficiently large. And for a constant J , the value of the function rises with the increase of $\text{SNR}_{1\%}$. Finally, we can calculated that $\eta_1 = -2.27$ of conventional system is smaller than $\eta_2 = -1.34$ of proposed system (negative values are the result of log-transformation in the function), which means our system have better comprehensive performance.

V. HARDWARE ARCHITECTURE

In this section, a general hardware architecture of proposed large-scale SCMA detector is proposed. Fig. 6 shows the overview of this architecture including three modules: *Control module*, *Buffer*, and *DMPA detecting module*. The design details are presented as follows.

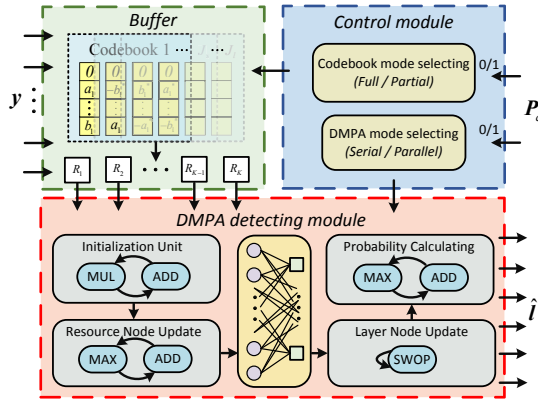


Fig. 6. General architecture of proposed large-scale SCMA detector.

1) *Control module*: This module is made up of two parts: the codebook mode selecting unit and the DMPA mode selecting unit. Both of the units have two modes to choose, and are represented as “0” and “1”. In codebook mode selecting unit, “0” means the conventional / rotation manner with codebook of all users (size J_l), and “1” means the extension manner with codebook of the first copy (size J_s). In DMPA mode selecting unit, “0” means serial decoding of each user, and “1” means parallel decoding of all users. Control signal P_c has the lengths of two bits to choose the preferred modes.

2) *Buffer*: This module is used to store the codebook of given SCMA system for the convenience of initialization procedure. It has a flexible memory space, which is determined by the input control signal. When mode “1” is selected, the buffer only needs to store J_s -size codebook (normally $J_s \ll J_l$), but user $J_s + 1 \sim J_l$ need to calculate their own codebook in the initialization unit thereafter.

3) *DMPA detecting module*: This module uses the deterministic message passing algorithm, whose details can be referred to [9, 10]. It chooses the serial or parallel procedure according to the input control signal. This module is made up of four parts: i) Initialization Unit: It calculates the initial belief of independent users with the received signal and specific codebook. The received signal consists of K complex resource nodes, thus the input is $2K$ -parallel. The codebook costs $2MNJ$ memory blocks of 8-bit length each in total.

ii) Resource Node Updating Network: It calculates the sum of belief and outputs the largest of the M^{d_f-1} intermediate beliefs in $(d_f - 1) \log_2^M$ steps of comparison with $M^{d_f-1} - 1$ buffers. Thus, each unit requires $M^{d_f} - M$ buffers, and the entire network requires $Kd_f(M^{d_f} - M)$. iii) Layer Node Updating Network: It swops the input R-to-L values to become new L-to-R values. Layer node beliefs would be reset when the received signals are detected. iv) Probability Calculating Unit: It finds the codeword with largest belief and map it back into binary information bits. Beliefs of M codeword are compared with each other in each unit. Hence, $\log_2^M$ steps of comparison with $M - 1$ buffers are required. And the total number of buffers consumed by the entire module would be $J(M - 1)$.

VI. CONCLUSION

A systematic codebook design of large-scale SCMA is proposed. Rotation and extension have been adopted. Hardware architecture has been proposed as well. Future work will be directed towards combination with other detectors [11].

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